

PE-EC802C — VLSI Design Automation

Semester VIII • 3L:0T:0P • 3 Credits

Module 4: VLSI Simulation & Logic Synthesis

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. VLSI Simulation Techniques

VLSI Simulation

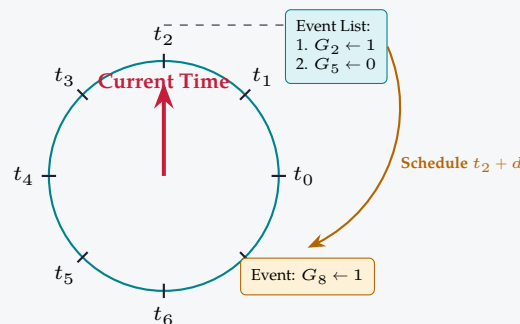
VLSI Simulation is the process of building a software model of a digital circuit and executing it with test input sequences to verify its functional correctness, check signal timings, and isolate design faults prior to physical fabrication.

1.1 Gate-Level Event-Driven Simulation

Event-driven simulators calculate logic outputs only when an input signal transition (an *event*) occurs. This is highly efficient because less than 10% of gates in a typical digital circuit switch states during any clock cycle.

- **Event Queue:** Stores pending signal updates sorted by execution time.
- **Time Wheel:** A circular queue scheduling events at future times. A pointer represents current simulation time (t). When the pointer moves to slot i , the events linked to t_i are evaluated. If a gate output changes, new events are scheduled at $t_i + d$ (where d is gate delay).

Time Wheel Scheduler Concept



1.2 Switch-Level Simulation

Models digital circuits at the transistor level by representing MOS transistors as voltage-controlled switches.

- **States:** Signals are modeled using more than two states, usually including 0 (low), 1 (high), X (unknown), and Z (high impedance).
- **Strengths:** Signals are assigned strengths to resolve bus contention (e.g. supply strength, pull strength, charge storage strength). An active pull-up transistor overrides a weak pull-down resistor.

2. Two-Level Logic Synthesis

Logic synthesis compiles high-level RTL descriptions (Verilog/VHDL) into an optimized gate-level netlist. Two-level logic synthesis targets sum-of-products (SOP) forms.

2.1 Quine-McCluskey (QM) Minimisation

An exact tabular method that guarantees finding the absolute minimum SOP expression.

1. Represent minterms in binary and group them by the number of 1s they contain.
2. Combine minterms that differ by exactly one bit (e.g., 0100 and 0101 combine to 010—), replacing the differing bit with a dash.
3. Repeat combinations until no more terms can be merged. The remaining uncombined terms are **Prime Implicants (PI)**.
4. Construct a Prime Implicant chart, identify **Essential Prime Implicants (EPI)** (columns containing only one checkmark), and select the minimum subset of PIs to cover the remaining minterms.

2.2 Espresso Heuristic Minimiser

Because QM suffers from exponential time complexity ($O(2^{2^n})$) and is unusable for more than 10 inputs, industrial synthesis tools use the Espresso heuristic optimizer. Key iterative steps in Espresso:

- **Expand:** Squeezes out variables by expanding cubes to cover adjacent minterms, reducing the total number of terms.
- **Irredundant:** Selects a minimum subset of prime cubes that covers the entire ON-set, deleting redundant terms.
- **Reduce:** Shrinks cubes to their smallest possible dimensions without uncovering any minterms, allowing the algorithm to search in new directions during the next Expand phase.

3. Binary Decision Diagrams (BDD)

Binary Decision Diagram

A **Binary Decision Diagram (BDD)** is a directed acyclic graph (DAG) representation of a Boolean function. It is constructed by recursively applying Shannon's expansion theorem.

3.1 Shannon's Expansion Theorem

Any Boolean function $f(x_1, x_2, \dots, x_n)$ can be expanded about variable x_1 as:

$$f = x_1 \cdot f_{x_1} + x_1' \cdot f_{x_1'} \quad (1)$$

where $f_{x_1} = f(1, x_2, \dots, x_n)$ is the positive cofactor, and $f_{x_1'} = f(0, x_2, \dots, x_n)$ is the negative cofactor.

Shannon's Expansion Example

Problem:

Expand the function $f = AB + B'C$ about variable A .

Solution:

1. Compute cofactors about A :

$$f_A = (1)B + B'C = B + C \quad (\text{using absorption } B + B'C = B + C)$$

$$f_{A'} = (0)B + B'C = B'C$$

2. Combine terms using Shannon's theorem:

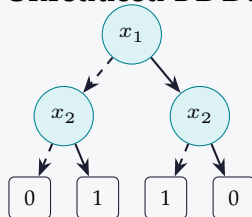
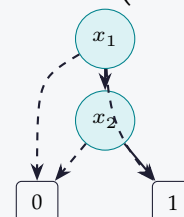
$$f = A \cdot (B + C) + A' \cdot (B'C)$$

3.2 Reduced Ordered BDDs (ROBDD)

An **Ordered BDD (OBDD)** is a BDD where variables are evaluated in a fixed order along every path from root to leaf. A **Reduced OBDD (ROBDD)** is obtained by applying three reduction rules:

1. **Merge Duplicate Leaves:** Combine all leaf nodes containing 0 into a single 0-terminal, and all 1-leaves into a single 1-terminal.
2. **Merge Duplicate Nodes:** If two internal nodes u and v represent the same variable and have identical low and high child nodes, merge them.
3. **Eliminate Redundant Nodes:** If an internal node has the same low and high child node w , delete the node and redirect all incoming edges directly to w .

BDD to ROBDD Reduction Steps

Unreduced BDD:

Reduced ROBDD (after merging):


ROBDDs are canonical representations of Boolean functions. For a fixed variable ordering, every Boolean function has exactly one unique ROBDD.

4. Quick Revision Page

4.1 Key Formulas & Syntheses

Logic Synthesis & BDD Formulations

1. Shannon's Expansion Theorem:

$$f(x_1, \dots, x_n) = x_1 \cdot f(1, x_2, \dots, x_n) + x_1' \cdot f(0, x_2, \dots, x_n)$$

This is the mathematical foundation for constructing BDD nodes.

2. Complexity bounds of exact QM minimization:

$$N_{PI} \approx \frac{3^n}{n} \quad (\text{worst-case number of prime implicants for } n \text{ variables})$$

3. Event scheduler delay equation:

$$t_{\text{schedule}} = t_{\text{current}} + d_{\text{gate}}$$

4.2 One-Line Definitions

- **Time Wheel:** A circular scheduler queue used in event-driven simulators to track gate updates.
- **Switch-Level:** Transistor-level logic simulation where nodes represent strengths and states.
- **Prime Implicant:** A product term (cube) that cannot be combined with any other term to form a simpler term.
- **EPI:** A prime implicant that covers at least one minterm not covered by any other prime implicant.
- **Espresso:** An industrial heuristic logic minimizer that runs in polynomial average time.
- **Cofactor:** The Boolean function obtained by setting a specific variable to 0 or 1.
- **ROBDD:** A canonical directed acyclic graph representing a Boolean function, minimized using reduction rules.
- **Canonical Form:** A unique representation of a mathematical object, allowing easy comparison.

4.3 Comparison: Simulators & BDDs

Gate-Level vs. Switch-Level Simulation:

Feature	Gate-Level Simulator	Switch-Level Simulator
Primitive Element	Boolean logic gates (AND, OR, XOR).	MOS transistors (nodes, switches).
Execution Speed	Very fast (evaluates only on logic changes).	Slow (calculates charge flows and transistor strengths).
Accuracy	Ideal logic behavior.	Captures bi-directional pass transistor behavior.

BDD vs. ROBDD:

Feature	Binary Decision Diagram (BDD)	Reduced Ordered BDD (ROBDD)
Ordering	No fixed order of variables along paths.	Strict, identical variable ordering along all paths.
Uniqueness	Not unique (multiple representations).	Canonical (unique representation).
Leaf Nodes	Contains duplicate 0 and 1 terminal leaves.	Contains exactly one 0 and one 1 terminal.

4.4 Frequently Asked Questions / Viva Questions

Exam Q&A

1. **Explain the advantage of event-driven simulation over compiler-driven simulation.**

Compiler-driven simulation evaluates every gate in the circuit during every time step, which is slow. Event-driven simulation only evaluates gates whose inputs undergo a transition (an event). Since only 2 – 10% of gates change states in a clock cycle, event-driven simulation is 10 times faster.

2. **Why are multiple strengths used in switch-level simulators?**

Transistors can have different dimensions, driving strengths, or functions (e.g. pull-up vs. pull-down). Using strengths allows the simulator to resolve contentions when multiple transistors drive the same bus line, ensuring that the stronger driver dominates.

3. **What is the difference between Quine-McCluskey (QM) and Espresso?**

QM is an exact minimization algorithm that guarantees the absolute minimum SOP form but has exponential time complexity. Espresso is a heuristic optimizer that uses Expand, Reduce, and Irredundant steps to find a near-optimal solution in polynomial average time, making it suitable for large circuits.

4. **Why is variable ordering critical in ROBDDs?**

The size (number of nodes) of an ROBDD is highly sensitive to the chosen variable order. A good ordering can result in a linear number of nodes ($O(n)$), while a poor ordering can make the node count grow exponentially ($O(2^n)$) for the exact same function.

5. **What are cofactors, and how are they used in logic synthesis?**

A cofactor is a simplified Boolean function obtained by setting a variable to a constant value ($f_{x_i} = f(x_i = 1)$). They are used in Shannon's expansion to recursively build BDDs and simplify logic expressions.

6. **How does the Espresso algorithm use the "Reduce" operation to escape local minima?**

The Reduce operation shrinks the size of prime cubes. This temporarily exposes some minterms, allowing the subsequent Expand step to expand cubes in different directions. This cycle helps the algorithm escape local minima.

7. **What is a canonical representation, and why are ROBDDs canonical?**

A canonical representation is a unique way of writing a mathematical object. ROBDDs are canonical because, for a fixed variable ordering, any Boolean function has exactly one unique ROBDD structure. This allows checking if two functions are identical in $O(1)$ constant time.

8. **Explain the "Merge Duplicate Leaves" reduction rule in BDDs.**

Frequently Asked / Viva Questions (contd.)

This rule combines all leaf nodes containing 0 into a single 0-terminal, and all leaves containing 1 into a single 1-terminal. This reduces the node count and creates a single converging point for all paths.

9. What is the function of the event queue in an event-driven simulator?

The event queue stores pending signal changes sorted by their execution times. As the simulator moves forward, it fetches events from the current time slot, evaluates the affected gates, and schedules new events at future slots.

10. Why do we need logic synthesis tools?

Logic synthesis automatically translates high-level functional code (Verilog/VHDL) into an optimized gate netlist. This speeds up design, guarantees logical correctness, and optimizes the gate count to meet timing and area constraints.